

Black-Litterman Model

For Portfolio Optimisation and Asset Allocation

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INTRODUCTION

The Black-Litterman Model is a mathematical model designed to optimize asset allocations in portfolio by incorporating investor's view with market equilibrium returns.

This model leverages on a Bayesian framework to combines subjective views, sentiment and risk appetite with the historical market information, thus overcomes the limitations of modern portfolio theory and traditional mean-variance optimisation (MVO) framework by leveraging on reverse optimisation method.

It was developed in 1990 by Fischer Black and Robert Litterman at Goldman Sachs thus deriving its name.

Mathematical Expression of Black-Litterman Model

$$E[R] = \left[(\tau \Sigma)^{-1} + P' \Omega^{-1} P \right]^{-1} \left[(\tau \Sigma)^{-1} \Pi + P' \Omega^{-1} Q \right]$$

Here,

E[R] : Adjusted (Posterior) Expected Return Vector

τ : Scalar

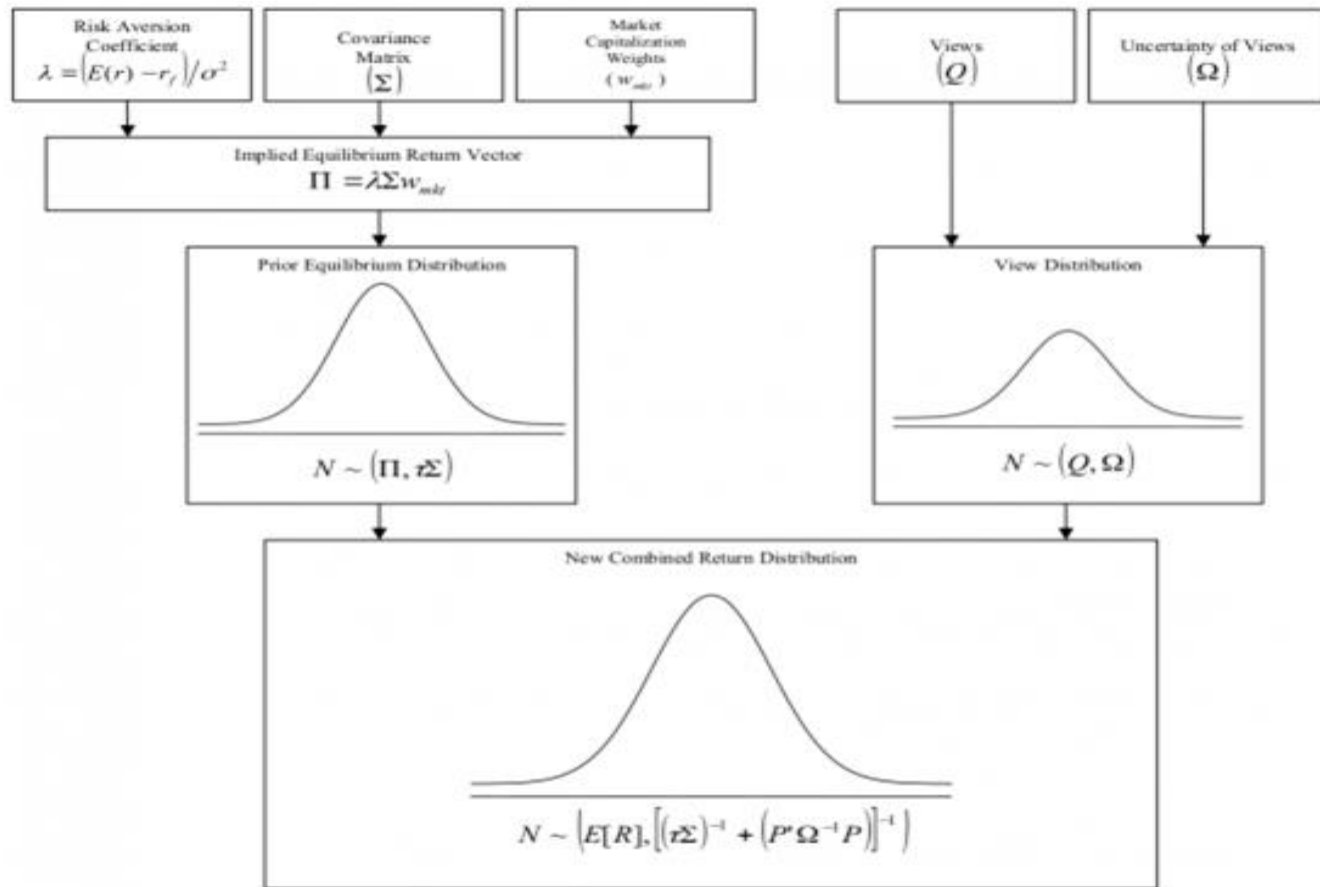
Σ : Covariance Matrix of excess returns

Π : Implied Equilibrium Return

P : View Matrix $(k \times n)$

Q : View Front/Vector $(k \times 1)$

Ω : Uncertainty Matrix (diagonal covariance matrix of error terms)



KEY TERMINOLOGIES FOR BLACK-LITTERMAN MODEL

- 1) Equilibrium Market Returns
- 2) Investor's View
- 3) Variance-Covariance Matrix (Σ)
- 4) View Matrix (P)
- 5) View Front (Q)
- 6) Implied Equilibrium Return (Π)
- 7) Tau (τ)
- 8) Uncertainty Matrix (Ω)

Equilibrium Market Returns

Equilibrium market returns represent the returns that are implied by the market's capitalization weights. These returns are typically derived from the Capital Asset Pricing Model (CAPM), which assumes that the market portfolio is mean-variance efficient.

$$E(r) = r_f + \beta r_m + \alpha$$

Implied Equilibrium Return (π)

The implied equilibrium return Π is the return vector that corresponds to the market equilibrium. It is calculated using the market capitalization weights and the covariance matrix of asset returns.

Implied Equilibrium Return is calculated using Reverse Optimisation Method:

$$\pi = \lambda \cdot \Sigma \cdot w_m$$

Here,

λ : [Risk Aversion](#) Parameter

Σ : Covariance Matrix

w_m : Market Capitalization Weights Vector

Variance-Covariance Matrix (Σ)

Variance-Covariance Matrix (also known as Covariance matrix) captures the pairwise covariances between the returns of the assets. In Black-Litterman model, we calculate it for excess returns of the assets.

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} & \sigma_{14} & \sigma_{15} & \sigma_{16} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} & \sigma_{24} & \sigma_{25} & \sigma_{26} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} & \sigma_{34} & \sigma_{35} & \sigma_{36} \\ \sigma_{41} & \sigma_{42} & \sigma_{43} & \sigma_{44} & \sigma_{45} & \sigma_{46} \\ \sigma_{51} & \sigma_{52} & \sigma_{53} & \sigma_{54} & \sigma_{55} & \sigma_{56} \\ \sigma_{61} & \sigma_{62} & \sigma_{63} & \sigma_{64} & \sigma_{65} & \sigma_{66} \end{bmatrix}$$

Covariance Matrix

$$\begin{bmatrix} \text{Var}(x_1) & \dots & \dots & \text{Cov}(x_n, x_1) \\ \vdots & \ddots & & \vdots \\ \text{Cov}(x_n, x_1) & \dots & \dots & \text{Var}(x_n) \end{bmatrix}$$

Investor's View

Investor views are the subjective opinions about the expected performance of certain assets. These views can be either absolute or relative and are dependent on investor's analysis and projections. K denotes number of views.

There are two types of views:

- 1) Absolute View: These views are independent of assets and their correlation. It specifies returns of particular asset

Eg. Stock A will give an return of 8%.

- 1) Relative View: These views consist of two or more assets and their projected performances with respect to each other.

Eg. If Stock C depreciated by 10% then Stock A and Stock B will increase.

Example Views

Suppose, there is a Portfolio named QFC, in which there are **5 assets**: Stock A, Stock B, Stock C, Stock D and Stock E

And views of a market expert are as follow:

- 1) Stock A will have expected return of 8% (Absolute view)
- 2) Asset B will outperform Asset C by 3% (Relative view)
- 3) Asset D will outperform Asset E by 2% (Relative view)

View Matrix (P)

The view matrix P is used to represent the investor's views in a structured format. Each row in the view matrix corresponds to a specific view and shows the weights of the assets involved in that view. The View Matrix gives a general idea of assets involved in the view. It is $k \times n$ matrix, where k is number of views and n is number of assets.

In View Matrix, the **rows** denotes different **views** while the **columns** denotes the different **assets** present in the portfolio.

Each row of the View Matrix (P) sums to one. The distribution of the assets can be done using [Satchell and Scrowncroft method](#) (equally weighted) or by market capitalization.

Example View Matrix

Using the previous example case of Portfolio QFC , we get View Matrix $(P)_{3 \times 5}$
As k is 3 and n is 5 here.

We get P as :

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 \end{bmatrix}$$

View Front/Vector (Q)

View Front (Q) gives the quantification of subjective view. The numerical values of the view is expressed using view vector.

It is a $k \times 1$ column matrix.

Example View Front (Q)

Using the previous example case of Portfolio QFC, we get View Vector $_{3 \times 1}$ as k here is 3.

$$Q = \begin{bmatrix} 0.08 \\ 0.03 \\ 0.02 \end{bmatrix}$$

Tau (τ)

It is a chosen scalar which shows the degree of confidence in the prior estimates of the equilibrium returns (π). In essence, it scales the covariance matrix of returns to indicate how much we trust the historical estimates of returns.

The parameter τ adjusts the weight given to the prior (historical data) versus the investor's views. A smaller τ implies less confidence in the equilibrium returns and more weight given to the investor's views.

Typically τ lies between 0 and 1 while in majority of cases $\tau = 0.025$ is used.

Uncertainty Matrix (Ω)

Ω represents the uncertainty (variance) associated with the investor's views. It is a diagonal matrix where each diagonal element corresponds to the variance of a particular view. It incorporates views along with the confidence.

$$\Omega = \text{diag}(P \cdot \tau \cdot \Sigma \cdot P^T)$$

$$\Omega = \begin{bmatrix} (p_1 \Sigma p_1') * \tau & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & (p_k \Sigma p_k') * \tau \end{bmatrix}$$

IMPLEMENTATION OF THE RESULTS

The Posterior Equilibrium Returns from the Black-Litterman Model is then used to calculate the optimal portfolio weights using Mean-Variance Optimisation (MVO) Method.

